Valence as an Informational Symmetry Problem: A Topological Model of Chemical Bonding Based on Harmonic Proportions

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Abstract

We propose an informational symmetry framework for atomic valence, where s (spherical/projection) and p (directional/confinement) orbitals represent orthogonal information channels. The **Harmonic Weight** P—defined as the irreducible sum of s:p electron counts—classifies main-group elements into three fundamental categories: (1) *Binary-closed* ($P = 2^n$) corresponding to inertness, (2) *Prime-tensioned* ($P \in \{3, 5, 7\}$) driving reactivity, and (3) *Structural* ($P = 2$) enabling covalent networks. The model successfully predicts periodic trends in Periods 2–3 without empirical fitting, suggesting that chemical bonding emerges from geometric constraints in an information-theoretic space.

1 Introduction

Periodic trends in main-group chemistry have been explained through various paradigms: octet rules , electronegativity , and VSEPR theory. While empirically successful, these approaches lack a unifying first-principles foundation. Here we propose that valence behavior emerges from *informational symmetry* conditions in the orbital space. We treat s and p orbitals as orthogonal information channels, with s-orbitals representing isotropic projection (global phase information) and p-orbitals encoding directional confinement (local phase gradients). Chemical stability then corresponds to *binary information closure*—states where the s:p ratio achieves minimal descriptive complexity in a binary basis.

2 Theoretical Framework

2.1 Information-Theoretic Foundation

Let $\mathcal{H}_s$ and $\mathcal{H}_p$ denote Hilbert spaces for s- and p-channel information, with dimensions N_s and N_p respectively. The total valence information is encoded in:

$$\mathcal{H}_{\text{valence}} = \mathcal{H}_s \otimes \mathcal{H}_p.$$

The entanglement entropy between channels:

$$S = -\text{Tr}(\rho_s \log \rho_s) = -\text{Tr}(\rho_p \log \rho_p),$$

quantifies information symmetry, with minimal S indicating maximal symmetry (inert states).

2.2 Harmonic Weight as Topological Invariant

For valence occupancy (N_s, N_p) with $\gcd(N_s, N_p) = d$, define reduced integers $a = N_s/d$, $b = N_p/d$. This defines a point in projective space $\mathbb{P}^1(\mathbb{Q})$. The **Harmonic Weight**:

$$P(a : b) = a + b \tag{1}$$

acts as a height function. Crucially:

- $P = 2^n$: Minimal height in binary basis $\rightarrow$ informationally closed
- P prime: Irreducible in binary basis $\rightarrow$ informationally tense
- $P = 2$: Self-dual point (1:1) $\rightarrow$ enables mutual saturation

Edge cases ($N_s = 0$ or $N_p = 0$) correspond to singly-connected limits outside the *sp*-closure hierarchy.

3 Methodology

For main-group *sp* shells ($N_s \in [0, 2]$, $N_p \in [0, 6]$):

1. Reduce $N_s : N_p$ to irreducible $a : b$ via gcd
2. Compute $P = a + b$
3. Classify: *Prime* vs. *Power of 2* vs. *Structural* ($P = 2$)
4. Compare to qualitative reactivity and periodic trends

4 Results: Periods 2 & 3

4.1 Classification Results

Table **??** shows the complete classification. Key patterns:

- **Inert closure**: Ne, Ar ($P = 4 = 2^2$) with 1:3 ratio
- **Structural axes**: C, Si ($P = 2 = 2^1$) with 1:1 ratio
- **Prime tension**: B/Al ($P = 3$), N/P ($P = 5$), O/S ($P = 3$), F/Cl ($P = 7$)

4.2 Interpretation

- **Binary closure ($P = 2^n$)**: Complete information cycles $\rightarrow$ chemical inertness
- **Prime tension (P prime)**: Irreducible asymmetry $\rightarrow$ bonding drive
- **Structural ($P = 2$)**: Perfect balance $\rightarrow$ covalent network formation

Table 1: Harmonic Weight classification for Periods 2–3 elements

Element Class	**Config.**	N_s:N_p	**Reduced**	P
B Prime	s^2p^1	2:1	2:1	3
C Structural	s^2p^2	2:2	1:1	2
N Prime	s^2p^3	2:3	2:3	5
O Prime	s^2p^4	2:4	1:2	3
F Prime	s^2p^5	2:5	2:5	7
Ne Binary-closed	s^2p^6	2:6	1:3	4
Al Prime	s^2p^1	2:1	2:1	3
Si Structural	s^2p^2	2:2	1:1	2
P Prime	s^2p^3	2:3	2:3	5
S Prime	s^2p^4	2:4	1:2	3
Cl Prime	s^2p^5	2:5	2:5	7
Ar Binary-closed	s^2p^6	2:6	1:3	4

5 Quantitative Extensions

Define valence fraction:

$$r_V = \frac{N_s}{N_s + N_p}, \tag{2}$$

which equals 1/4 for closed 1:3 patterns. We hypothesize:

- r_V correlates with first ionization energy within periods
- Electronegativity scales inversely with P for prime-weight elements
- Bond strengths maximize for complementary prime pairs

6 Predictions and Falsifiability

6.1 Materials Implications

1. **Doping**: Prime-weight dopants in Group IV semiconductors should show enhanced carrier generation

2. **Catalysis**: Transition metal complexes with prime-weight ligand configurations may exhibit superior activity
3. **Superconductivity**: Binary-closed configurations might suppress, while prime configurations enhance superconducting T_c

6.2 Experimental Tests

- Measure carrier generation rates for different dopant P values
- Correlate catalytic turnover frequencies with ligand P_{eff}
- Test P-based predictions under high pressure/field conditions

6.3 Falsifiability Conditions

The model fails if:

- Strong d-orbital participation doesn't break P-predictions
- Relativistic effects don't scramble the prime/binary hierarchy
- High-pressure phases follow different P-renormalization

7 Limitations and Future Directions

7.1 Current Limitations

- Neglects electron correlation beyond mean-field
- Ignores relativistic and spin-orbit effects
- Static picture without dynamics
- Orbital orthogonality approximation

7.2 Future Extensions

- Ternary $N_s : N_p : N_d$ generalization with $P_3 = a + b + c$
- Connection to topological quantum numbers
- Time-dependent information flow
- Solid-state k-space information geometry

8 Conclusion

The Harmonic Weight P provides a simple yet powerful classification of main-group valence behavior based on informational symmetry principles. By treating chemical bonding as an emergent phenomenon from information-theoretic constraints, we offer a new perspective on periodic trends with predictive power for materials design. The framework bridges quantum chemistry, information theory, and materials science, opening avenues for first-principles design of functional materials.

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